

ATLEE AGCM, AB, Canada

WMO: 719860

Lat: 50.810N

Lon: 111.006W

Elev: 2608

StdP: 13.36

Time Zone: -7.00 (NAM)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.6	-15.7	-26.3	1.4	-20.5	-20.8	1.9	-15.6	33.3	28.8	29.4	30.4	6.7	170	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	88.2	61.3	84.6	60.8	81.0	60.2	65.4	78.7	63.9	77.5	62.3	76.3	12.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	89.0	70.4	59.1	82.5	68.4	57.2	76.9	67.1	31.7	78.2	30.6	77.4	29.3	76.2	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.1	25.7	22.4	DB	-28.4	94.2	4.4	3.3	-31.5	96.6	-34.1	98.5	-36.5	100.4	-39.7	102.8	(4)
(5)			WB	-28.6	69.4	4.3	2.4	-31.7	71.1	-34.2	72.5	-36.7	73.9	-39.8	75.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	15.7	13.0	26.9	40.8	52.7	59.8	65.7	64.4	55.4	41.3	26.1	13.7		
	DBStd	22.09	14.55	14.89	14.49	8.86	7.78	6.14	5.59	6.18	9.17	9.50	13.66	14.04		
	HDD50	5365	1064	1036	718	294	60	5	0	0	47	296	720	1125		
	HDD65	9407	1529	1456	1183	726	386	177	60	89	308	736	1167	1590		
	CDD50	1634	0	0	1	18	144	299	487	447	210	25	3	0		
	CDD65	201	0	0	0	0	5	22	83	70	21	0	0	0		
	CDH74	2996	0	0	0	8	159	325	1140	963	395	6	0	0		
	CDH80	954	0	0	0	0	28	78	389	322	137	0	0	0		
	WSAvg	10.9	10.7	10.1	11.5	12.4	11.7	11.1	10.4	9.9	10.4	11.2	10.8	10.1		
	PrecAvg	13.9	0.5	0.3	0.6	1.1	1.8	3.2	1.7	1.5	1.3	0.7	0.6	0.5		
	PrecMax	20.8	1.3	0.9	1.6	3.2	4.1	8.0	4.8	4.0	3.3	1.7	1.7	1.3		
(15) Precipitation	PrecMin	5.9	0.1	0.0	0.0	0.0	0.3	1.1	0.2	0.1	0.2	0.1	0.0	0.1		
	PrecStd	3.7	0.4	0.3	0.4	0.7	1.2	1.7	1.3	0.9	0.9	0.4	0.4	0.3		
	DB	47.5	46.3	61.9	73.5	84.0	87.5	93.1	91.8	89.8	73.6	63.4	45.3			
	MCWB	38.8	38.4	46.3	51.5	57.6	60.9	63.2	61.1	59.6	55.3	47.9	39.0			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	41.8	40.8	56.3	67.6	78.5	81.6	88.3	87.6	84.3	66.6	54.5	40.2			
	2% MCWB	36.4	35.2	43.7	48.8	55.5	59.9	62.8	61.1	58.3	51.5	43.5	35.8			
	5% DB	38.7	36.7	51.2	62.2	74.0	77.5	84.3	83.5	78.0	61.6	48.3	37.3			
	5% MCWB	34.9	33.3	41.3	45.8	53.3	59.2	62.2	60.9	56.7	48.6	40.2	33.5			
	10% DB	35.6	33.1	44.8	57.2	69.3	73.8	80.4	79.5	72.9	56.5	42.7	33.2			
	10% MCWB	32.9	30.7	37.9	43.2	51.6	57.3	61.6	60.2	54.9	45.8	36.9	30.1			
	0.4% WB	40.3	40.0	47.6	52.1	58.8	65.9	68.9	66.7	62.5	56.1	48.6	39.9			
	0.4% MCDB	46.5	45.6	59.8	70.3	80.1	77.9	83.5	79.0	79.0	70.5	62.7	43.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.8	35.7	44.9	49.4	56.6	63.1	65.9	64.9	59.9	52.6	44.4	36.0			
	2% MCDB	41.2	39.8	55.1	66.0	74.9	76.0	79.3	78.8	77.8	64.7	52.9	39.9			
	5% WB	35.1	33.5	41.3	46.5	54.8	60.9	64.5	63.2	57.9	49.3	40.5	33.6			
	5% MCDB	38.5	36.7	49.8	59.5	70.6	73.4	77.4	77.6	75.4	59.7	47.0	37.6			
	10% WB	32.8	30.6	38.0	44.1	52.8	58.8	63.1	61.5	55.8	46.6	37.6	30.2			
	10% MCDB	35.6	32.9	44.4	55.4	66.3	70.7	76.7	75.7	71.9	55.6	42.1	33.1			
	MDBR	16.7	17.6	18.9	22.1	24.0	21.9	24.4	24.1	24.0	21.5	18.0	15.9			
	5% DB	17.1	20.2	26.8	31.5	31.2	27.3	30.8	31.4	33.3	28.9	24.7	19.1			
(36) Mean Daily Temperature Range	MCWBR	13.2	15.7	15.5	15.6	13.4	11.4	11.0	11.5	13.4	15.4	15.2	15.3			
	MCDBR	17.3	19.4	25.5	29.3	27.4	24.6	25.4	26.0	30.8	27.3	23.7	19.3			
	MCWBR	13.8	15.4	15.5	15.3	12.3	12.0	11.2	10.9	13.1	14.8	15.4	15.5			
	MCDBR	17.1	20.2	26.8	31.5	31.2	27.3	30.8	31.4	33.3	28.9	24.7	19.1			
(40) Clear-Sky Solar Irradiance	taub	0.266	0.282	0.293	0.349	0.375	0.367	0.364	0.385	0.333	0.292	0.263	0.256			
	taud	2.375	2.401	2.457	2.302	2.273	2.356	2.388	2.296	2.428	2.519	2.485	2.382			
	Ebn at Noon	247	273	292	283	279	282	281	268	272	268	250	233			
	Edh at Noon	19	25	28	37	40	37	36	37	29	22	17	16			
(44) All-Sky Solar Radiation	RadAvg	374	693	1131	1558	1860	1968	2130	1727	1242	757	412	287			
	RadStd	38	62	80	115	123	118	99	115	106	85	35	27			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page